

Homework:

Malware Classification based on *Federated Learning*

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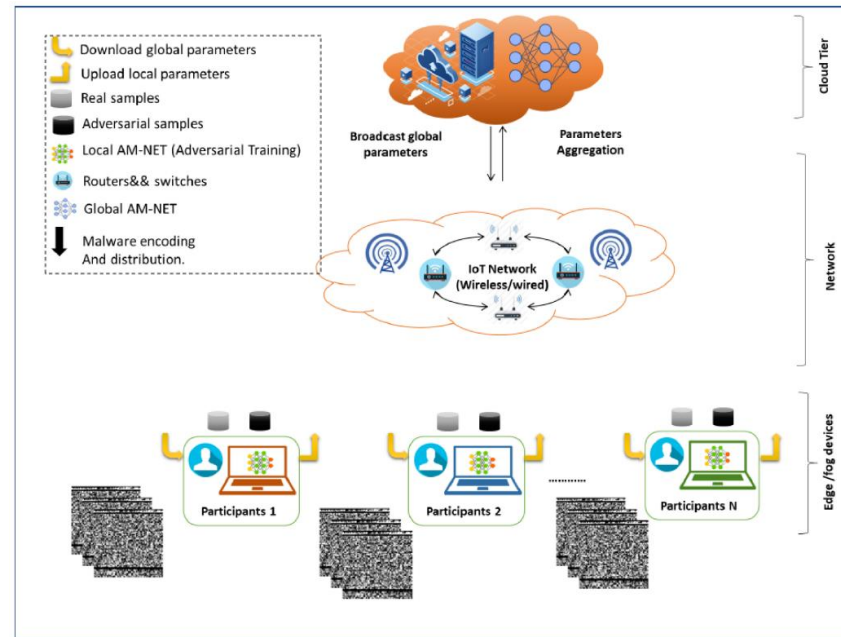
- **Replace** the machine learning method in Midterm **with Federated Learning**.
- Some References (focus only on their aspects related to federated learning, disregard the other parts):
 - [M. Abdel-Basset, et al. "Efficient and Lightweight Convolutional Networks for IoT Malware Detection: A Federated Learning Approach." *IEEE Internet of Things Journal* 10.8 \(2022\): 7164-7173.](#)
 - No need for its adversarial training in this homework.
 - [S. Shukla, et al. *Malware Detection using Federated Learning based on HPC Events and Localized Image features*. 2019.](#)
 - [S. B. Atitallah, Maha Driss, and Henda Ben Ghezala. "FedMicro-IDA: A federated learning and microservices-based framework for IoT data analytics." *Internet of Things* 23 \(2023\): 100845.](#)

Hints

1. Repeat Midterm's job (Step 3. Hybrid Analysis), but replace the machine learning method with Federated Learning.

- You can divide it into 3 clients.
- Some libraries for Federated Learning. I would suggest using **TensorFlow Federated**, because the following homework may use related resources. You could also try the **FLOWER**.
 - <https://www.tensorflow.org/federated>
 - [Top 7 Open-Source Frameworks for Federated Learning - www.apheris.com](https://www.apheris.com)
 - [Flower: A Friendly Federated Learning Framework](#)

2. Training and testing by machine learning model for detecting malware.



Upload the result to Moodle

- Upload your code and the ***detailed*** reports to Moodle.
 - Code: You must add some comments in your code for easy understanding.
 - Report Files: Document (Word)
 - Zipped to a file. The file name should be “Team Name_HW5.zip”